

1 Introduction

The Standard and Poor's 500, or simply the S&P 500 is a stock market index that tracks the performance of 500 large companies in the US. It is widely followed and seen as a representation of the US stock market, with an average annual return of 9.8% since its inception in 1926. The index includes companies that have increased dividends for 25 consecutive years, known as the S&P 500 Dividend Aristocrats. It is capitalization-weighted, with the top 10 companies accounting for 21.8% of the index's performance. Funds that track the index are recommended by prominent investors like Warren Buffett for long-term investors. Although it currently only includes US-based companies, multinational companies are also included. The index is used in forecasting the economy through the Conference Board Leading Economic Index and has various ticker symbols like GSPC, INX, and \$SPX. It is updated frequently throughout the trading day by S&P Dow Jones Indices.

1.1 The Objective of the Study

The objective of the study is to apply various classification methods, namely Logistic Regression, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and K-Nearest Neighbors (KNN) - to the weekly S&P stock market data. The aim is to analyze the results derived from these methods and identify which one yields the highest or maximum prediction accuracy.

1.2 Background Information

This data set consists of percentage returns for the stock index over 1089 weeks for 21 years, from the beginning of 1990 to the end of 2010. For each data, we have recorded the percentage returns for each of the first previous trading weeks, Lag1 through Lag5. We have also recorded Volume (the number of shares traded on the previous day, in billions), Today (the percentage return on the date in question) and Direction (whether the market was Up or Down on this date). The structure of the data set can be seen in the below.

1.3 Data Visualization

1.3.1 Correlation Analysis

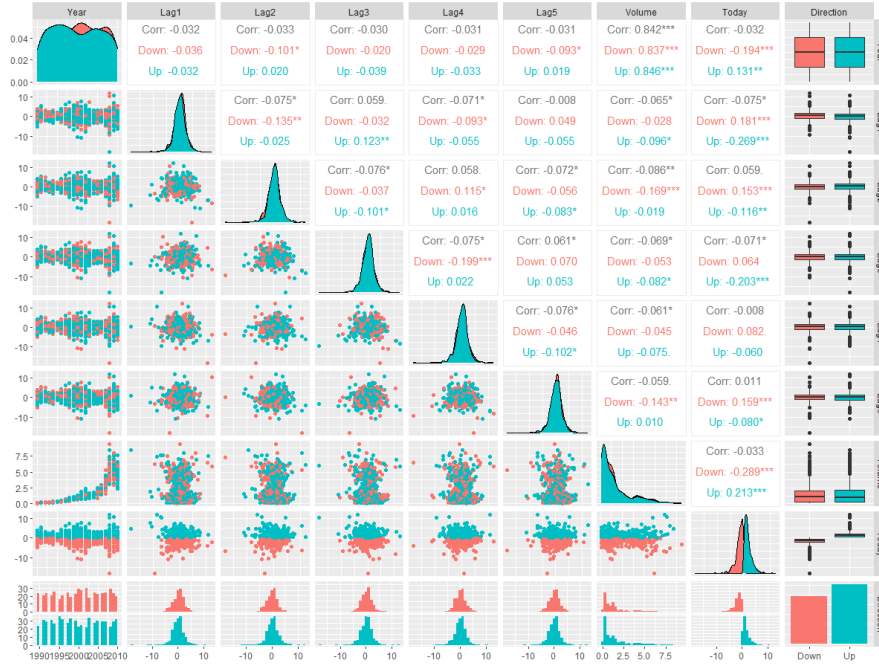


Figure 1: Plot 1

	Year	Lag1	Lag2	Lag3	Lag4	Lag5	Volume	Today
Year	1.00000000	-0.032289274	-0.03339001	-0.03000649	-0.031127923	-0.030519101	0.84194162	-0.03245989
Lag1	-0.03228927	1.000000000	-0.07485305	0.05863568	-0.071273876	-0.008183096	-0.06495131	-0.0750318
Lag2	-0.03339001	-0.074853051	1.00000000	-0.07572091	0.058381535	-0.072499482	-0.08551314	0.0591667
Lag3	-0.03000649	0.058635682	-0.07572091	1.00000000	-0.075395865	0.060657175	-0.06928771	-0.0712436
Lag4	-0.03112792	-0.071273876	0.05838153	-0.07539587	1.000000000	-0.075675027	-0.06107462	-0.0078258
Lag5	-0.03051910	-0.008183096	-0.07249948	0.06065717	-0.075675027	1.000000000	-0.05851741	0.0110126
Volume	0.84194162	-0.064951313	-0.08551314	-0.06928771	-0.061074617	-0.058517414	1.00000000	-0.0330777
Today	-0.03245989	-0.075031842	0.05916672	-0.07124364	-0.007825873	0.011012698	-0.03307778	1.0000000

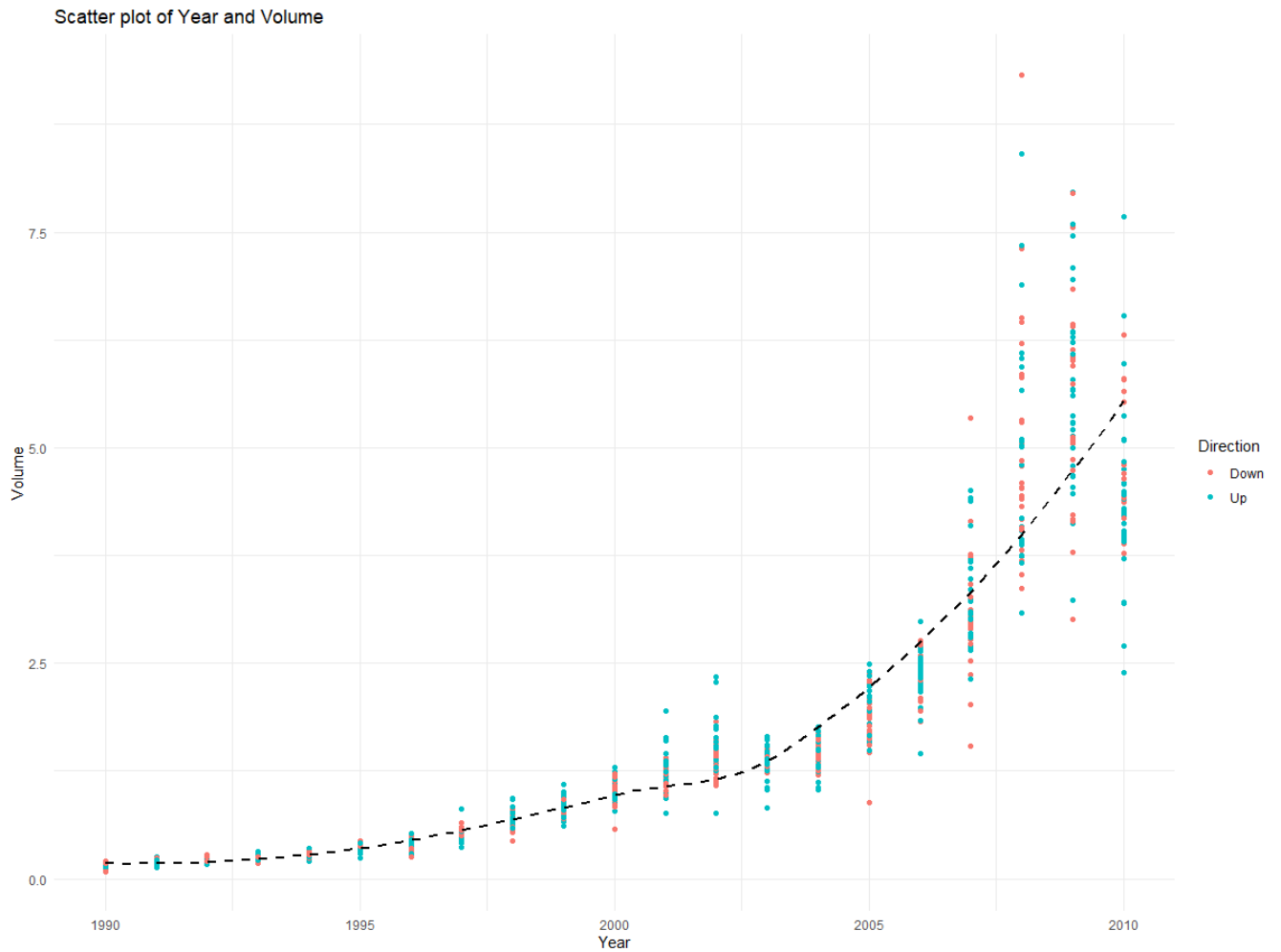


Figure 2: Scatter Plot of Volume and Year

From the summary statistics and correlation statistics obtained to study the relation between variables and from visualization it is clear that there is high correlation between Volume and Year. This appears to be an exponential relationship where volume increases exponentially as a function of year. Some years seem to have more or less variations than others. Looking at the shape of the various Lag features and the Year. There does seem to be some auto correlation in the variability of the Lag and the year. Perhaps some years people are more skittish than other years, and this takes a while to wear off. There appears to be very little if any correlation between Lag and other lags. Direction appears slightly skewed by a few of the lags, perhaps Lag5, and Lag1.

1.4 Full Data

1.4.1 Logistic Regression (All Predictors)

Call :

```
glm(formula=Direction~Lag1+Lag2+Lag3+Lag4+Lag5+Volume,family=binomial,data=Weekly)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.6949	-1.2565	0.9913	1.0849	1.4579

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.26686	0.08593	3.106	0.0019 **
Lag1	-0.04127	0.02641	-1.563	0.1181
Lag2	0.05844	0.02686	2.175	0.0296 *
Lag3	-0.01606	0.02666	-0.602	0.5469
Lag4	-0.02779	0.02646	-1.050	0.2937
Lag5	-0.01447	0.02638	-0.549	0.5833
Volume	-0.02274	0.03690	-0.616	0.5377

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1496.2 on 1088 degrees of freedom

Residual deviance: 1486.4 on 1082 degrees of freedom

AIC: 1500.4

Number of Fisher Scoring iterations: 4

From the variables and summary statistics in the logistic regression model output generated, we can see variables Intercept and Lag2 are statistically significant which means their beta coefficients are not equal to zero.while Lag1 also has a p-value close to 0.05, suggesting it might be marginally significant.

1.5 Reduced Data

Now we reduced the data, by randomly splitting the data set into 80% for training and 20% for testing to compare the performance of each methods. First we fit the all the classification methods using a training data with Lag2 as the only predictor. Next, we then fit all the classification methods using training data with different combination of predictors Lag1, Lag2 interaction Lag1*Lag2 and also analysis the confusion matrix of each method.

1.5.1 Confusion Matrix Analysis

From analyzing the confusion matrix result of each the classification methods, similarly they all tells us that the model does fit the data particularly well. The "Up" direction is guessed most of the time. Most of the mistakes come from guessing that the market is going to go "Up" when it really is going to go "Down".

1.5.2 Performances of Methods with One Predictor

Table 1: Performance Comparison of Classification Methods

Method	Overall Fraction of Correct Prediction (%)
Logistic Regression (LR)	54.59
Linear Discriminant Analysis (LDA)	54.59
Quadratic Discriminant Analysis (QDA)	55.04
k-Nearest Neighbors (KNN) with $K = 1$	53.67
k-Nearest Neighbors (KNN) with $K = 10$	55.96
k-Nearest Neighbors (KNN) with $K = 15$	56.68

It appears as though 54.59% of the responses are predicted correctly (model accuracy). In other words, 45.41% is the test error rate, which is often overly optimistic. for both logistic regression and LDA.

It appears as though 55.04% of the responses are predicted correctly (model accuracy). In other words, 44.96% is the test error rate for QDA.

It appears as though 53.67% of the responses are predicted correctly (model accuracy). In other words, 46.33% is the test error rate for KNN, K=1

It appears as though 55.67% of the responses are predicted correctly (model accuracy). In other words, 44.33% is the test error rate for KNN, K=10.

It appears as though 56.68% of the responses are predicted correctly (model accuracy). In other words, 43.32% is the test error rate for KNN, K=15.

If we compare the model accuracy (overall fraction of correct prediction), we observe that Logistic Regression and Linear Discriminant Analysis (LDA) have similar accuracies, followed by Quadratic Discriminant Analysis (QDA). However, KNN with K=15 exhibits the highest accuracy, followed by KNN with K=10 and KNN with K=1. Therefore, KNN with K=15 provides the best results among the model.

1.5.3 Performances of Methods with(Different Combinations of Predictors)

Table 2: Performance Comparison of Classification Methods

Method	Overall Fraction of Correct Prediction (%)
Logistic Regression (LR)	55.04
Linear Discriminant Analysis (LDA)	54.59
Quadratic Discriminant Analysis (QDA)	53.67
k-Nearest Neighbors (KNN) with $K = 1$	55.50
k-Nearest Neighbors (KNN) with $K = 10$	56.88
k-Nearest Neighbors (KNN) with $K = 15$	51.83

We fit the LR model using the training data, with predictors Lag1,Lag2 and Lag1*Lag2 as the interaction. the percentage of correct predictions on the test data is 55.04%. In other words, 44.96% is the test error rate.

We fit the LDA model using the training data, with predictors Lag1,Lag2 and Lag1*Lag2 as the interaction. the percentage of correct predictions on the test data is 54.49%. In other words, 45.51% is the test error rate.

We fit the QDA model using the training data, with predictors Lag1, Lag2 and Lag1*Lag2 as the interaction. the percentage of correct predictions on the test data is 53.67%. In other words, 46.33% is the test error rate.

We fit the KNN, k=1 model using the training data, with predictors Lag1, Lag2 and Lag1*Lag2 as the interaction. the percentage of correct predictions on the test data is 55.50%. In other words, 44.50% is the test error rate.

We fit the KNN, k=10 model using the training data, with predictors Lag1, Lag2 and Lag1*Lag2 as the interaction. the percentage of correct predictions on the test data is 56.88%. In other words, 43.12% is the test error rate.

We fit the KNN, k=15 model using the training data, with predictors Lag1, Lag2 and Lag1*Lag2 as the interaction. the percentage of correct predictions on the test data is 51.83%. In other words, 48.17% is the test error rate.

If we compare the model accuracy (overall fraction of correct prediction), we observe that k-Nearest Neighbors (KNN) with K=10 demonstrates the highest accuracy at 56.88%, followed by Logistic Regression at 55.04%, and KNN with K=1 at 55.50%. Linear Discriminant Analysis (LDA) follows closely with an accuracy of 54.59%, while Quadratic Discriminant Analysis (QDA) has the lowest accuracy at 53.67%. Notably, KNN with K=15 shows the lowest accuracy at 51.8%. Therefore, among the models listed, KNN with K=10 provides the best results in terms of accuracy.

1.6 Conclusion

In our analysis, we evaluated the performance of the classification methods, namely K-Nearest Neighbors (KNN), Logistic Regression, Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis (QDA).

Our findings revealed that the KNN model consistently outperformed the other methods in terms of prediction accuracy. This superior performance was observed across different values of 'k' (the number of neighbors) and for various combinations of predictors.

When we reduced the dataset to one predictor, the KNN model with $k=15$ provided the highest model accuracy, as indicated by the overall fraction of correct predictions. When we used different combinations of predictors on the reduced data, the KNN model still demonstrated the best performance. In these cases, $k=10$ was identified as the optimal value, delivering high prediction accuracy with an optimized model.

An important insight from our study is the effect of increasing 'k' in the KNN model. As 'k' increases, the model becomes more generalized, which reduces the risk of over-fitting but may lead to less accurate predictions for individual data points.

While a general rule of thumb suggests setting the value of 'k' as $\sqrt{N}/2$ (where 'N' is the number of samples in the training dataset), this value may not always be optimal for every dataset. Therefore, it is recommended to experiment with different values of 'k' to find the best value for a specific dataset.

In conclusion, our study found that the KNN model demonstrated superior performance. However, the optimal configuration of the model (specifically the value of 'k') can vary depending on the specific characteristics of the dataset. Therefore, thorough testing and validation are crucial when applying these models to new datasets.